Vinayak Kamat

AI/ML Engineer | Computer Vision, Generative Models & LLM Agents

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Summary

AI/ML Engineer with 1+ year of production experience building computer-vision and LLM-powered systems. Trained PyTorch detection models to ~85% accuracy, optimized inference with OpenCV and ONNX, and built retrieval-augmented generation pipelines grounded in domain data. Hands-on with generative pipelines, LLM agents, function calling, and structured output extraction. Comfortable taking a model from research idea to production — measuring quality, catching failures, and iterating. Active open-source contributor across vision, document AI, and on-device LLM inference.

Skills

Computer Vision & Generative: PyTorch, OpenCV, Transformers, image detection & classification, inference optimization (ONNX), generative pipelines, model calibration

LLMs & Agents: LangChain, LangGraph, LlamaIndex, RAG pipelines, retrieval over catalogue/domain data, function calling, structured JSON outputs, prompt engineering (few-shot, CoT, ReAct)

Programming: Python (primary), C++, JavaScript, SQL, FastAPI, Flask, REST APIs

MLOps & Cloud: MLflow, Docker, Git, AWS (Lambda, S3, ECR, DynamoDB), CI/CD, Datadog, model monitoring, drift detection

Data: Pandas, NumPy, dataset preparation

Experience

KAN Innovations, AI/ML Engineer — Mumbai, Maharashtra
May 2025 – present
1 year 7 months

- Built and evaluated production RAG pipelines using LangChain/LlamaIndex and Chroma, grounding LLM outputs in retrieved domain data and validating quality on every release
- Implemented function calling and structured JSON extraction from documents, with schema validation and fallback handling for unreliable outputs
- Deployed ML services on AWS (Lambda, ECR, DynamoDB); optimized PyTorch-to-ONNX inference to reduce latency for production models
- Tracked experiments, model versions, and quality metrics with MLflow; monitored production behavior with Datadog

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Built a PyTorch-based image detection model achieving ~85% accuracy, from dataset preparation through training and evaluation
- Improved an existing posture/eye detection model through data augmentation and training refinements using OpenCV

Projects

nanoMind — On-Device LLM Inference
Offline LLM app running open models on constrained hardware — generative pipeline engineering end to end.

- Ran GGUF generative models on Android via llama.cpp with ARM-specific optimizations and real-time token streaming
- Public repo: github.com/vinayakkamatcodes/nanoMind

Aethelgard — LLM Quality Gateway
Real-time gateway that scores LLM outputs for hallucination and risk, keeping generative outputs reliable in production.

- Built a FastAPI gateway using a secondary judge model to score generative outputs, with metrics streamed to Datadog
- Designed fallback logic so low-confidence outputs degrade safely
- Public repo: github.com/vinayakkamatcodes/Aethelgard

ocrify — Document Extraction Pipeline
OCR-driven extraction pipeline applying computer vision and structured parsing to scanned documents.

- Built an OCR-driven extraction tool (Dart) — computer-vision preprocessing plus structured output parsing
- Public repo: github.com/vinayakkamatcodes/ocrify

Financial AI Agent (Demo)
Prototype agent for catalogue-style data retrieval and reasoning over financial data.

- Built modular retrieval and reasoning pipelines with query routing and response refinement
- Demonstrates grounding LLM agents in a business's own structured data

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified GATE (Graduate Aptitude Test in Engineering)

Active open-source contributor — github.com/vinayakkamatcodes

Participated in Kaggle competitions focused on applied machine learning